



Development and experimental evaluation of a low-power LoRa wearable sensor node for livestock monitoring

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ABSTRACT

This paper presents the development of an energy-efficient wearable sensor platform for livestock activity monitoring applications. The proposed device integrates an ESP32 microcontroller, an ADXL345 triaxial accelerometer, and an SX1278 LoRa transceiver into a compact neck-mounted enclosure optimized for prolonged field deployment. A dedicated power management strategy, combining low-dropout regulation and deep sleep modes, enables approximately 104 h of autonomy on a single 18,650 Li-ion battery. A gateway based on the STM32L433 microcontroller ensures reliable data acquisition and seamless logging for real-time analysis. Experimental trials conducted in both rural and urban environments confirm stable communication. However, they also highlight that hardware selection has a decisive influence on system performance: in the present implementation, the SX1278 limited the communication range to around 200 meters. A web-based dashboard was additionally developed for sensor data visualization and historical monitoring. Overall, this work demonstrates that beyond detection algorithms, careful hardware design and component selection are critical factors for achieving robust, energy-efficient, and field-deployable livestock monitoring solutions.

1. Introduction

Estrus detection is one of the most critical parameters for ensuring effective reproductive management in dairy cattle. Accurate and timely identification of estrus not only optimizes artificial insemination timing but also helps maintain herd productivity by reducing the interval between calvings. Globally, dairy cattle and buffalo account for approximately 96% of milk production—81% from cattle and 15% from buffalo—while other species such as goats, sheep, and camels contribute only 4% [1]. Ensuring efficient reproduction within these populations is therefore essential to sustaining productivity and long-term global food security.

Over the past decades, numerous studies have been conducted on estrus detection, exploring methods ranging from behavioral observation and hormonal assays to modern sensor-based approaches using accelerometers, temperature probes, infrared thermography, video analytics, and audio signal processing. These works have significantly advanced the accuracy and reliability of estrus monitoring systems. However, one crucial aspect remains underexplored in this field: the electronic and communication design of such devices. While detection algorithms have been extensively discussed, far fewer studies have

addressed the communication range of each model, the selection of sensing hardware, and the choice of wireless communication technology—all of which are essential for real-world deployment in large-scale farming environments.

Several researchers have investigated alternative approaches to livestock tracking, providing partial insights into energy consumption, communication capabilities, and sensing modalities. Wang et al. [2] tested a triaxial accelerometer (ADXL345) for behavioral tracking, reporting accurate detection but highlighting limitations in battery autonomy (48 h) and communication range (35 m). Higaki et al. [3] proposed a background image subtraction technique for tie-stall video surveillance, offering a non-invasive solution but requiring fixed infrastructure.

Different LoRa transceivers have been employed in related studies, reflecting trade-offs between energy efficiency, range, and hardware complexity. The RFM95 module has been used in several applications, including environmental gas sensing [4] and cattle activity observation [5], due to its low cost and ease of integration. The SX1278 module has also been widely adopted, particularly in accelerometer-based activity recognition systems such as those proposed by Bagwari et al. [6] and Li et al. [7], where its robustness and communication range were

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advantageous. More advanced modules, such as the SX1280, were tested by Hu et al. [8], enabling extended features but requiring higher configuration complexity. In contrast, other studies, such as those by Aloo et al. [9], Wang et al. [10], and Higaki et al. [11], focused primarily on sensing and classification approaches without explicitly reporting the LoRa transceiver used. This variety demonstrates the flexibility of LoRa technology, which can be adapted to different monitoring objectives and system constraints.

Beyond accelerometers, other sensing modalities have also been explored to expand livestock monitoring capabilities. Sakar et al. [12] combined sensor data with meteorological measurements to study cow behavior, while Marquez et al. [13] used infrared thermography and behavioral biometrics to detect skin temperature variations associated with estrus onset. Wang et al. [14] introduced an audio-based system capable of identifying both estrus status and individual cows from voiceprints. Wearable devices have also been proposed: Pérez et al. [15] evaluated the CowManager Sensor an ear-tag device integrating a 3D accelerometer, which collects behavioral data every minute and transmits aggregated results hourly via a wireless router. Similarly, Aloo et al. [16] applied machine learning techniques to accelerometer data from Kenyan dairy herds, though device autonomy was limited to approximately two days.

Overall, accelerometers remain the most common sensing solution in LoRa-based livestock monitoring platforms. Studies by Nyamuryekung et al. [17], Li et al. [18], and all relied on accelerometer-based nodes for behavioral tracking. Wang et al. [19] further demonstrated that triaxial accelerometers (ADXL345) can achieve high recognition accuracy, although energy constraints remain a challenge. Other sensing modalities have also been integrated, such as CO₂ and CH₄ gas sensors for environmental monitoring [4], and video-based surveillance with background subtraction [3]. This diversity illustrates that LoRa-enabled devices are not limited to motion tracking but can also support physiological, environmental, and multimodal sensing, broadening their applicability across precision livestock farming contexts.

In this paper, we focus not on re-evaluating estrus detection algorithms, but on addressing the electronic engineering challenges behind such systems. We present the development of a wireless embedded device specifically designed for estrus detection in dairy cows, combining a triaxial accelerometer with a LoRa-based long-range communication module. The design emphasizes low power consumption, extended battery life, portability, and reliable data transmission over large agricultural areas. Real-world experiments were carried out to validate detection performance, assess communication reliability, and evaluate the feasibility of long-term deployment.

The main contributions of this work are as follows:

- Development of a cost-effective wearable inertial sensor for estrus detection, incorporating energy-efficient mechanisms to achieve a battery lifetime of up to 48 h.
- Implementation of an energy optimization strategy based on configurable sleep/wake cycles, experimentally validated through multimeter measurements, with a consumption of ~11 mA in sleep mode and 52–53 mA in active mode.
- Design of a complete open-architecture prototype integrating multi-parameter sensing and LoRa-based communication, validated in rural and urban environments through analysis of communication range, hardware performance, and data quality using RSSI and SNR metrics.

This article is organized as follows: [Section 2](#) details the hardware design, power management, and wireless communication architecture. [Section 3](#) presents experimental results and compares them with existing systems from the literature. Finally, [Section 4](#) provides the conclusion and outlines perspectives for future improvements, including solar energy integration and device miniaturization.

2. Related work

Estrus detection in livestock is a critical component for improving reproductive efficiency and herd management. Over the past decades, various technologies and methodologies have been explored to monitor and accurately identify estrus events in animals. These approaches encompass sensor-based detection systems, communication and data transmission technologies, machine learning techniques, and power management strategies to ensure continuous and reliable operation of monitoring devices. In this section, we provide a comprehensive review of existing studies and applications in the field of estrus detection, highlighting the systems employed, the methods of data acquisition and transmission, the use of artificial intelligence for pattern recognition, and the strategies for power supply and energy management in sensor nodes. This review establishes the foundational context for our work and identifies opportunities for improvement in the development of efficient, intelligent, and autonomous estrus detection systems.

2.1. Estrus detection systems

2.1.1. Infrared thermography

Estrus in farm animals is marked by behavioral and physiological changes influenced by estrogen. The primary sign is standing to be mounted, while secondary signs include restlessness, vocalization, frequent urination, vulvar swelling, and a rise in body temperature [1]. Monitoring these indicators helps determine the optimal insemination time—about 12 h after standing heat in cows and 24 h in buffaloes due to delayed ovulation. These observations are crucial for maximizing conception rates.

Infrared Thermography (IRT) has emerged as a promising non-invasive technology for estrus detection in cattle and buffaloes. Recent studies have demonstrated its effectiveness: Marquez et al. [20] highlighted that automated IRT platforms could serve as an alternative to visual estrus detection in dairy cows. Tiwari et al. [21,21] reported that IRT, using a FLIR E60 infrared thermograph camera to monitor the muzzle and vulva, can efficiently detect estrus and its different stages in Sahiwal cows. The study was conducted on ten healthy, cyclic Sahiwal cows under standard herd management practices, with thermal images analyzed using FLIR Tools software to assess temperature variations at different estrus stages. Similarly, Marquez et al. [13] conducted their study on 18 primiparous Holstein cows at the University of Alberta's Dairy Research and Technology Centre, using infrared thermography in combination with behavioral monitoring and pipeline milking system observations. Ovarian status was tracked via transrectal ultrasonography (ALOKA SSD-500 scanner), and thermal measurements were collected throughout proestrus, estrus, ovulation, and post-ovulation stages. This integrated approach of IRT, behavioral monitoring, and reproductive ultrasonography significantly improved the accuracy of estrus detection. These studies collectively demonstrate that IRT, especially when integrated with behavioral monitoring, is a valuable tool for timely and accurate estrus detection in livestock.

2.1.2. Accelerometer and pedometers

Several commercial estrus detection systems rely on accelerometers integrated into different wearable platforms for dairy and beef cattle. Ear-tag-based systems, such as HeatTime [22] and CowManager [23], use accelerometers to monitor physical activity, resting behavior, and in some cases rumination. Neck-collar-mounted devices, including Nedap [24] and SenseHub Beef (Allflex Livestock Intelligence) [25], primarily employ accelerometers to track activity variations associated with estrus, while some systems also incorporate additional parameters [26] such as feeding behavior [27] or body temperature. Other solutions, such as Gyuho, utilize leg-mounted accelerometers, enabling accurate detection of locomotion changes. In addition to standalone accelerometer systems, some products integrate activity sensors with other sensing technologies, including ruminoreticular biocapsules like LiveCare [28],

which combine activity and body temperature measurements, and thermometric boluses such as SmartStock [29], mainly used for estrus and calving detection. Finally, ear-tag biosensors like Moocall HEAT [30,31] integrate accelerometer-based activity monitoring to enhance reproductive event detection.

2.1.3. Rumination and feeding behavior estrus detection

Rumination and feeding behavior are widely recognized as reliable indicators of estrus in dairy cows due to their strong association with physiological and behavioral changes during the reproductive cycle. As shown by measurements obtained from wearable sensors, significant reductions in rumination time were observed during estrus compared to non-estrus periods. Data collected using HR Tag sensors [32] revealed a decrease in rumination from 32.96 ± 0.54 min/2 h in non-estrus cows to 20.47 ± 2.68 min/2 h during estrus ($p < 0.01$) [33]. Similarly, CowManager Sensor data indicated a marked reduction in rumination time [33], from 22.96 ± 0.57 min/h in non-estrus to 12.90 ± 1.07 min/h during estrus ($p < 0.01$). In contrast, feeding time showed an opposite trend, with estrus cows exhibiting a significant increase in feeding activity (16.93 ± 0.99 min/h) compared to non-estrus cows (8.93 ± 0.65 min/h, $p < 0.01$) [33]. These results confirm that changes in rumination and feeding behavior provide strong discriminatory features for estrus detection and can be effectively exploited in sensor-based and AI-driven monitoring systems.

2.1.4. Rumen/Reticular boluses

Rumen or reticular bolus-based systems represent an effective alternative approach for estrus detection by enabling continuous monitoring of internal physiological parameters. The pioneering work by Burnett et al. [34] introduced the use of weighted rumen–reticular temperature boluses combined with collar-mounted activity sensors, demonstrating that increases in reticulo-rumen temperature, synchronized with elevated activity levels, are strongly associated with estrus and ovulation events. Subsequent studies have expanded this approach by integrating artificial intelligence and machine learning techniques for bolus-derived data analysis, as highlighted in the review by Hajnal et al. [35], which identified 16 research studies published between 2016 and 2022 focusing on intelligent cattle monitoring. More recently, Andrade et al. [36] employed multi-sensor boluses capable of simultaneously recording reticulo-rumen temperature and accelerometer-based activity at high temporal resolution, with wireless data transmission to cloud platforms. These systems confirm the high potential of rumen boluses for accurate, continuous, and environment-independent estrus detection, particularly when combined with advanced data analytics.

2.1.5. Camera- and vision-based estrus detection systems

Camera-based estrus detection systems provide a non-contact and cost-effective alternative to traditional visual observation by enabling continuous monitoring of cow behavior, particularly during nighttime when up to 68% of mounting events occur [37]. These systems rely on video recordings analyzed using computer vision and artificial intelligence techniques to automatically identify estrus-related behaviors such as mounting, standing heat, changes in posture, optical flow variations, and movement trajectories. By allowing a single camera to monitor multiple animals simultaneously, this approach significantly reduces labor demands and cost per animal while improving detection consistency. Complementary studies have combined behavioral scoring methods with imaging technologies, as demonstrated by [38], who integrated estrus behavior scores with endoscopic imaging to characterize cervical mucus changes during estrus. Furthermore, advanced imaging modalities such as infrared cameras have been employed to capture physiological changes associated with estrus; Wang et al. [39] utilized high-resolution thermal imaging of the eye and vulva to detect temperature variations linked to estrus events. Collectively, these studies demonstrate that camera-based systems—especially when integrated with AI-driven image analysis offer a promising, scalable, and

animal-friendly solution for accurate estrus detection.

2.2. Communication technologies for estrus monitoring

Reliable communication technologies are essential for real-time estrus monitoring systems, enabling continuous data transmission from on-animal sensors to farmers and decision-support platforms. In [40], an IoT-based wireless monitoring system was developed using a wearable ankle-mounted microcontroller to record hourly step counts and body temperature variations associated with estrus behavior. The system transmitted data wirelessly via Wi-Fi to a web-based interface, allowing real-time monitoring, automatic alerts, and remote access through the internet. This low-cost and energy-efficient solution demonstrated high reliability for estrus detection and timely insemination decisions. In a second approach [41], a low-power multiprotocol wireless communication architecture was implemented using a CC1352R-based control board, which supports Bluetooth 5.2 Low Energy (BLE). This system enabled bidirectional communication for sensor data acquisition and remote command transmission, highlighting the importance of ultra-low power wireless protocols for long-term livestock monitoring applications. Additionally, Hector J. et al. [42] employed RFID-based communication integrated with 3-axis accelerometer ear-tag sensors (CowManager™) to collect and transmit behavioral data such as activity and rumination. The RFID-enabled ear tags ensured reliable identification of individual animals while enabling continuous behavioral monitoring across the herd.

Nisit Pukrongta et al. [43] proposed a Low-Power Wide Area Network (LPWAN)-based cattle collar system for estrus monitoring, aiming to improve estrus detection accuracy through continuous behavioral data collection. The proposed collar integrates motion sensing using a gyroscope module (MPU-9250) and sound sensing via a microphone sensor, with data processing handled by a Seeeduno microcontroller. The refined sensor data are transmitted using LoRa (RFM95) technology operating at 925 MHz, enabling long-range, low-power wireless communication. A receiver unit based on an ESP32 forwards the collected data to a cloud platform (Google Sheets) for real-time storage and analysis. The system demonstrated reliable data transmission at five-second intervals and provided valuable insights into cattle movement patterns and vocalization behavior. This study highlights the effectiveness of LoRa-based LPWAN communication for scalable, energy-efficient estrus monitoring in precision livestock farming.

2.3. Artificial intelligence and machine learning for estrus detection

Recent advances in artificial intelligence (AI) and machine learning (ML) have significantly enhanced the automation, accuracy, and reliability of estrus detection systems, particularly through computer vision and deep learning techniques. These approaches reduce reliance on manual observation and enable continuous, non-invasive monitoring of animal behavior under real farm conditions. İbrahim Arıkan et al. [44] proposed a multi-stage deep learning framework for estrus detection based on mounting behavior analysis. Their approach combines convolutional neural networks (CNN/VGG) for initial mounting detection with YOLO-based object detection to localize the region of interest (ROI) where the mounting action occurs. The cropped ROI is subsequently analyzed using YOLO to identify individual cows, allowing simultaneous estrus detection and cow identification within a single automated pipeline.

Similarly, Su Myat NOE et al. [45] introduced a deep learning-based system for estrus detection through mating posture recognition using Mask R-CNN instance segmentation. Their method accurately segments individual cattle in complex farm environments and detects mounting events, followed by a lightweight tracking algorithm to improve temporal consistency. The proposed model achieved a high detection accuracy of 95.5% mAP, even with a relatively small training dataset, highlighting the robustness and practicality of vision-based deep

learning approaches. These approaches enhance pregnancy probability prediction and support optimal artificial insemination timing, thereby improving reproductive efficiency in dairy cattle.

3. Materials and methods

This section details the methodology and experimental setup adopted in this work, starting with an overview of the system architecture.

3.1. System architecture overview

The proposed system was entirely designed and developed in-house by the authors. The proposed system is a wireless sensor platform designed to detect estrus in cattle by analyzing motion data from a triaxial accelerometer. It consists of a low-power sensor node mounted on the cow's neck and a gateway connected to a personal computer for data collection. Communication between the two nodes is established via the LoRa protocol. The sensor node operates on battery power and periodically transmits data to the gateway, which stores the information in an Excel file using a Python-based logging script, as illustrated in Fig. 1. In addition, the collected data are visualized through a dedicated dashboard developed with PHP, JavaScript, HTML, CSS, and MySQL, where all recorded values are preserved and displayed for real-time monitoring of cattle movements.

For the experimental procedure, the system was evaluated in a controlled farm environment with a testing area of approximately 50 m × 30 m. Three cows were used in this study, and a single wearable node was sequentially attached to each cow for a monitoring period of three days per animal. After each period, the node was detached and transferred to the next cow in a rotating manner to ensure consistent data collection under similar environmental conditions. During the experiments, the sensor node operated at a sampling frequency of 1 Hz (one sample per second). The collected data were grouped using a 1-minute sliding window before transmission to reduce communication load and optimize energy consumption. The experiments focused on real outdoor conditions within the farm, including normal animal movement and environmental variability, in order to evaluate system functionality, communication reliability, and long-term stability of the proposed hardware platform.

3.2. Hardware description

The hardware implementation of the proposed system is divided into two main units: the wearable sensor node mounted on the animal and the gateway responsible for data reception. Their descriptions are provided below.

3.2.1. Design and implementation of the sensor node

The sensor node was engineered to be compact, lightweight, and suitable for long-term deployment on the neck of dairy cattle, ensuring minimal disruption to animal behavior. At its core, the system relies on an ESP32 microcontroller, a dual-core 32-bit processor with built-in Wi-Fi, Bluetooth, and low-power modes, including deep sleep, which significantly reduces energy consumption during standby periods. This feature allows the node to operate autonomously for several days on a single battery charge.

As shown in Fig. 3, the wearable sensor node is attached to the cow using a collar-mounted configuration. The node incorporates a rechargeable Li-Ion battery and an integrated TP4056 charging module. When the battery level becomes low, the collar can be temporarily detached from the animal, and the battery is recharged directly through the TP4056 charging circuit without removing it from the device enclosure. Once recharged, the collar is reattached to the animal for continued monitoring. To ensure communication reliability, the gateway continuously monitors incoming data packets. If no packet is received from a sensor node for three consecutive transmission intervals, an alert notification is automatically generated and sent to the farmer, indicating a potential communication or device issue.

Fig. 3 illustrates the complete wearable sensor node developed for livestock monitoring applications. The node integrates several hardware components, including (1) an ESP32 microcontroller responsible for data acquisition, local processing, and wireless communication management; (2) an MPU6050 accelerometer and gyroscope sensor used for motion monitoring; (3) an SX1278 LoRa transceiver module enabling long-range wireless communication; (4) a rechargeable 3.7 V 18,650 Li-ion battery serving as the main power source; (5) an AMS1117–3.3 V regulator; (6) TP4056 charging module for safe battery recharging; and (7) a power switch for device control. These components are integrated into a compact enclosure attached to a collar, providing a portable and low-power solution suitable for continuous livestock monitoring.

The SX1278 (RA-02) LoRa module was selected primarily due to its

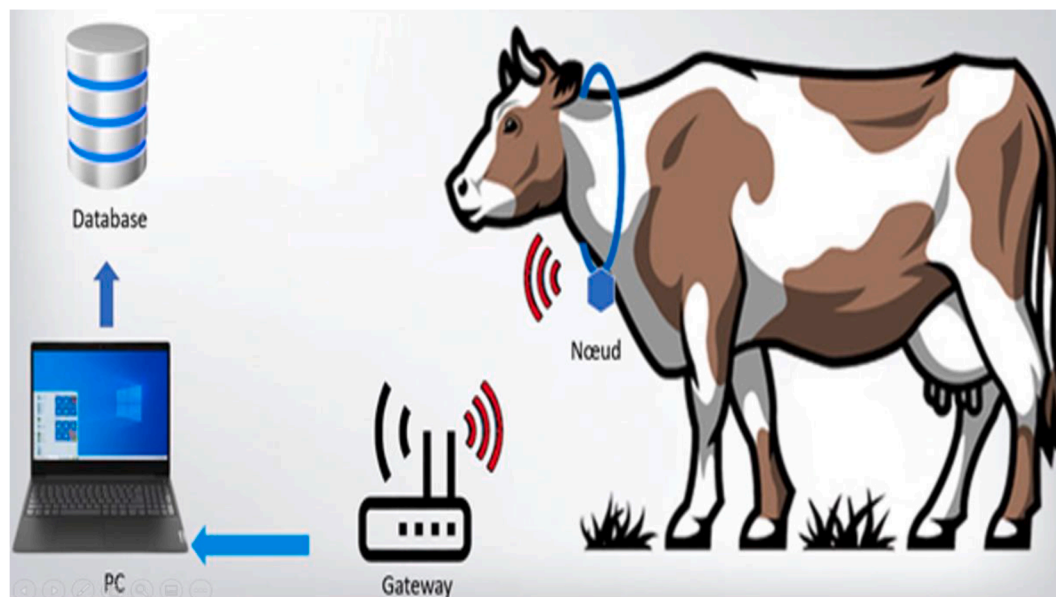


Fig. 1. System architecture and data flow.

wide availability on the market and its very low cost, making it highly suitable for large-scale livestock monitoring applications where minimizing the cost per deployed sensor node is essential. In addition, it provides reliable performance for short- to medium-range communication under our experimental conditions while maintaining low power consumption, which is critical for battery-operated wearable devices. The use of the 433 MHz frequency band also ensured stable and robust communication in the tested environment. Overall, this choice reflects a practical trade-off between cost, energy efficiency, and adequate communication performance for the targeted application.

The system also integrates an ADXL345 triaxial accelerometer for continuous motion monitoring, enabling the capture of fine-grained variations in posture and activity relevant to behavioral analysis. Sensor data are periodically sampled and transmitted at predefined intervals, ensuring an optimal balance between temporal resolution and energy efficiency, which supports long-term autonomous operation of the wearable device.

The system is powered by a 3.7 V 18,650 lithium-ion battery, selected for its high energy density and durability under field conditions. A low-dropout (LDO) voltage regulator is used to reduce the battery output to 3.3 V, providing stable power to the ESP32 and other components. To further optimize energy consumption, the microcontroller is programmed to enter deep sleep mode between transmissions, greatly extending the node's autonomy.

The wearable device was designed with particular attention to mechanical robustness, animal comfort, and long-term field usability. A custom 3D-printed enclosure was developed to securely integrate all electronic components within a compact collar-mounted structure, as illustrated in Fig. 4. The enclosure dimensions are approximately 90 mm × 70 mm × 25 mm, providing sufficient internal space for the electronic modules while maintaining a compact and lightweight form factor suitable for animal wearability. To improve durability under outdoor farm conditions, the enclosure includes a dedicated sealing interface to enhance water and dust resistance. In addition, two circular mounting openings were incorporated into the mechanical design to allow secure attachment of the device directly onto the animal collar, ensuring stable fixation during movement. The enclosure also provides mechanical protection against environmental factors such as rain, mud, and physical impacts, thereby improving long-term operational reliability.

Particular attention was also given to the electrical safety and operational stability of the wearable system during continuous use under animal movement conditions. To prevent circuit stress and

battery-related issues, a regulated power management stage was integrated to maintain a stable supply voltage to all components, even under dynamic load variations. This design protects the electronic circuitry from voltage fluctuations and ensures reliable long-term operation in real farm environments.

All electronic components are housed in a custom-designed enclosure mounted on the animal's neck. This enclosure is designed to be weather-resistant, shock-resistant, and comfortable enough for continuous wear during grazing and movement. The complete sensor node, integrating the ESP32, LoRa module, accelerometer, and battery inside a neck-mounted enclosure, is illustrated in Fig. 2, which shows the node design in Proteus, and in Fig. 3, which presents the implemented prototype for on-animal deployment.

3.2.2. Design and implementation of the gateway

The gateway was fully designed and implemented as part of this work to serve as the dedicated receiving and data logging unit for the proposed wearable sensor system, rather than relying on any commercial off-the-shelf gateway solutions. It is based on the STM32L433 microcontroller, a low-power Cortex-M4 MCU selected for its energy efficiency and suitability for embedded IoT applications. The microcontroller is interfaced with an SX1278 LoRa transceiver module, identical to that used in the sensor node, enabling reliable wireless communication under the same hardware configuration.

Upon reception of data packets from the sensor node, the gateway forwards the information to a personal computer (PC) via a UART-to-USB interface. A dedicated Python script running on the PC handles data decoding, extraction of relevant fields (including timestamps and triaxial acceleration measurements), and structured storage in an Excel (.xlsx) file for further processing and analysis.

This approach ensures that all behavioral data are archived chronologically and are directly compatible with post-processing tools, including machine learning algorithms for estrus detection.

To ensure stand-alone operation in off-grid or semi-mobile deployments, the gateway is powered by a dual-battery system, composed of two 3.7 V Li-ion cells connected to a 5 V step-up voltage regulator. This configuration provides sufficient and stable power to support both the STM32 and the LoRa module for extended periods. The choice of components and power strategy prioritizes energy efficiency and hardware simplicity, while maintaining compatibility with open-source software and readily available batteries.

Overall, the gateway acts as a central bridge between the sensor node

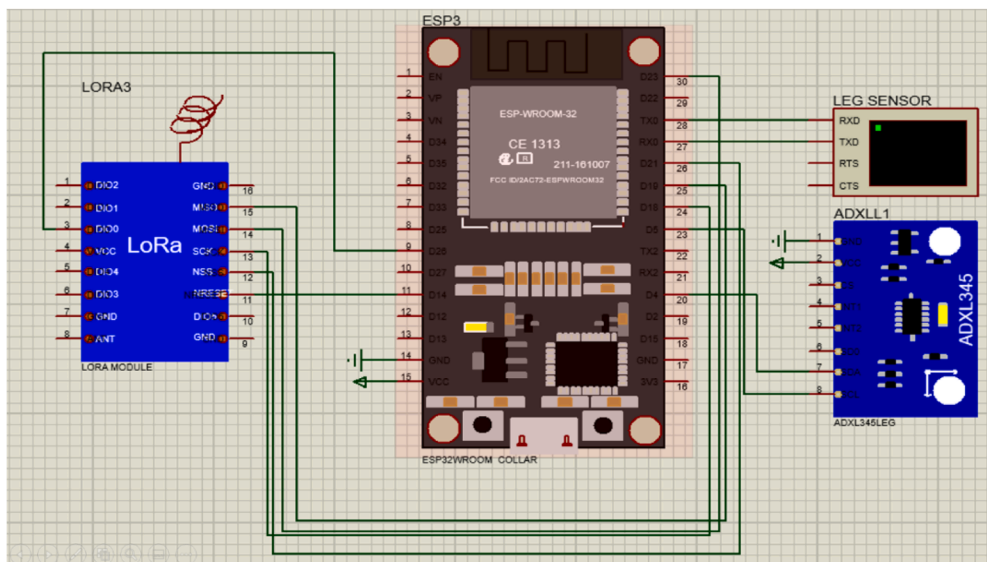


Fig. 2. Sensor node schematic designed in Proteus.

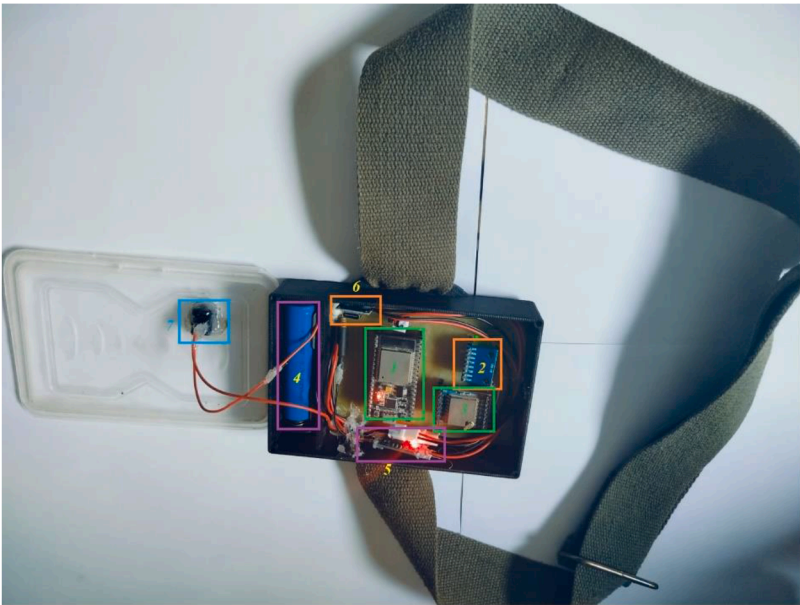


Fig. 3. Hardware implementation of the wearable IoT sensor node used in the proposed system.

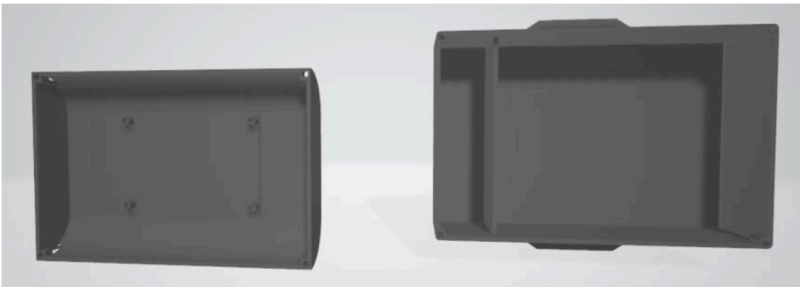


Fig. 4. 3D design of the wearable sensor node.

and the data analysis platform, ensuring low-latency communication, secure data capture, and scalability for future integration into multi-

node architectures or cloud-based livestock monitoring systems. The complete gateway setup is illustrated in Fig. 5, which shows the

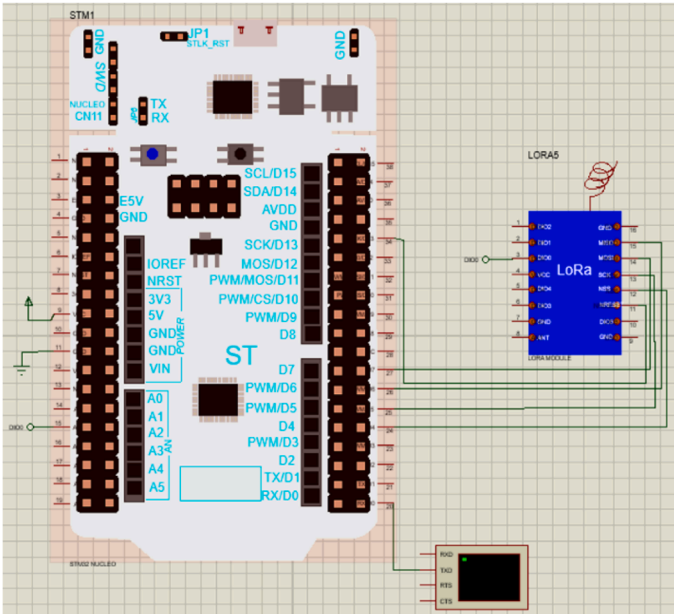


Fig. 5. Gateway schematic designed in Proteus.

hardware design, and in Fig. 6, which presents the prototype implementation highlighting the integration of the STM32, LoRa transceiver, and USB communication.

3.3. Power management and communication parameters

This section addresses the power management strategy and the communication parameters of the proposed system. It is divided into two parts: The first point discusses the power supply design, whereas the second point examines battery life estimation.

3.4. Power supply design

Several power supply configurations were tested in order to provide autonomous operation for the sensor node and gateway. The first setup used a USB power source, which allowed initial testing of the system. In a second configuration, the sensor was attached to the cow and powered by two batteries in combination with a 7805 linear regulator. This setup was intended to ensure sufficient voltage stability for the node. However, due to the significant heat dissipation observed with the 7805 regulator, an additional configuration was tested using a single 3.7 V 18,650 lithium-ion battery together with an AMS1117–3.3 V regulator, with the system supplied directly at 3.3 V. This regulator maintained a steady output voltage to power the ESP32-based sensor node. A schematic of the implemented AMS1117–3.3 V regulator module is presented in Fig. 7.

3.5. Battery life estimation

The sensor node operates in three distinct phases during each duty cycle: a sensing phase, during which the ESP32 microcontroller is active and acquires data from the MPU-6050 accelerometer; a transmission phase, during which the LoRa SX1278 module transmits a data packet; and a sleep phase, during which the system enters low-power mode between operations. Current consumption in each phase was measured experimentally using a calibrated digital multimeter. The measured current values are summarized as follows:

- Sleep mode (I Sleep mode current: $I_{\text{sleep}} \approx 11$ mA, comprising the ESP32 light-sleep draw (~ 2 mA) and the quiescent current of the AMS1117 linear voltage regulator.

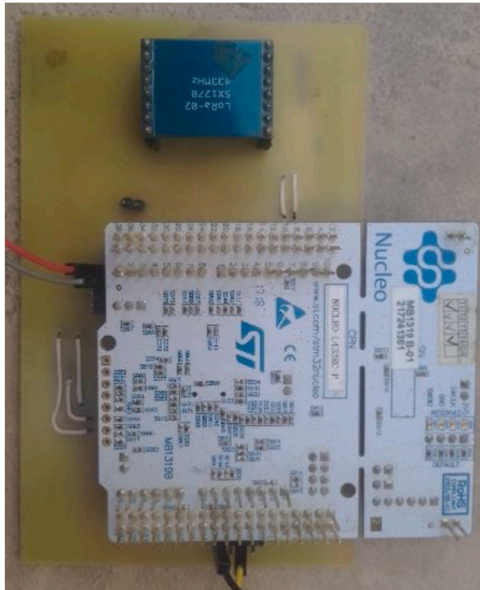


Fig. 6. Implemented gateway prototype.

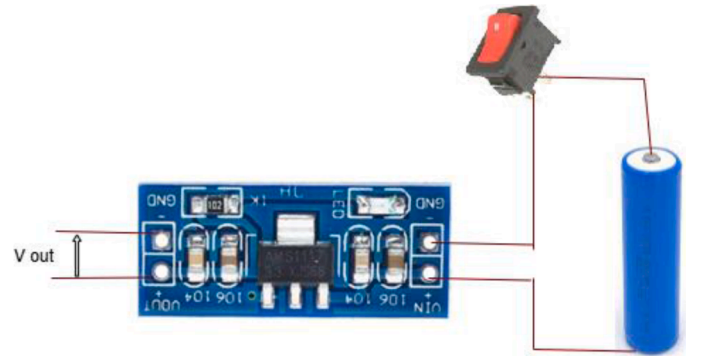


Fig. 7. Power supply schematic based on a 3.7 V 18,650 lithium-ion battery and an AMS1117–3.3 V regulator for the ESP32 sensor node.

- Sensing phase current (ESP32 active, MPU-6050 reading): $I_{\text{sense}} \approx 2\text{--}5$ mA, depending on the processing load of the microcontroller.
- LoRa transmission peak current: $I_{\text{TX}} \approx 52\text{--}53$ mA, measured during the LoRa SX1278 RF burst.

To evaluate the impact of transmission interval configuration on system performance, several data aggregation window sizes were experimentally tested, including 10 s, 30 s, 1 min, 10 min, and 30 min. The results indicate that short-time windows (10 s, 30 s, and 1 min) preserve high-resolution motion dynamics and ensure accurate representation of animal activity patterns. Among these configurations, the 1-minute window provides the best compromise between temporal resolution and energy efficiency, while maintaining sufficient detail for reliable behavioral monitoring.

In contrast, longer aggregation windows (10 min and 30 min) significantly reduce the number of transmissions, leading to lower energy consumption. However, this reduction comes at the cost of information loss, as short-duration behavioral events and rapid motion changes may not be captured effectively. This trade-off is critical in wearable IoT systems, where balancing data fidelity and battery lifetime is essential.

The battery lifetime of the sensor node can be approximated using the classical capacity-over-current model:

$$T_{\text{life}} = \frac{C_{\text{batt}}}{I_{\text{avg}}} \quad (1)$$

where C_{batt} represents the usable battery capacity (Ah), and I_{avg} denotes the average current consumption of the system (A). In this work, a 2200 mAh lithium-ion battery ($C_{\text{batt}} = 2.2$ Ah) is used to power the sensor node. To experimentally evaluate the power consumption, measurements were carried out using a digital multimeter under different operating conditions. Two representative operating states were considered: the sleep mode, where the system operates in a low-power state, and the normal operating mode, where both sensing and LoRa transmission are active. The experimental setup is illustrated in Figs. 8 and 9, showing the current measurement in both operating conditions. (Table 1).

The average current over a full duty cycle of period T_{cycle} is expressed as:

$$I_{\text{avg}} = \frac{I_{\text{read}} \cdot T_{\text{read}} + I_{\text{TX}} \cdot T_{\text{TX}} + I_{\text{sleep}} \cdot T_{\text{sleep}}}{T_{\text{cycle}}} \quad (2)$$

where I_{read} , I_{TX} , and I_{sleep} are the typical currents in each phase, and T_{cycle} is the total cycle duration.

With a $C_{\text{batt}} = 2.6$ Ah Li-ion cell, the theoretical autonomy of the node is:

$$T_{\text{life}} = \frac{2.2}{0.021} \approx 104\text{h} \approx 4 \text{ days} \quad (3)$$

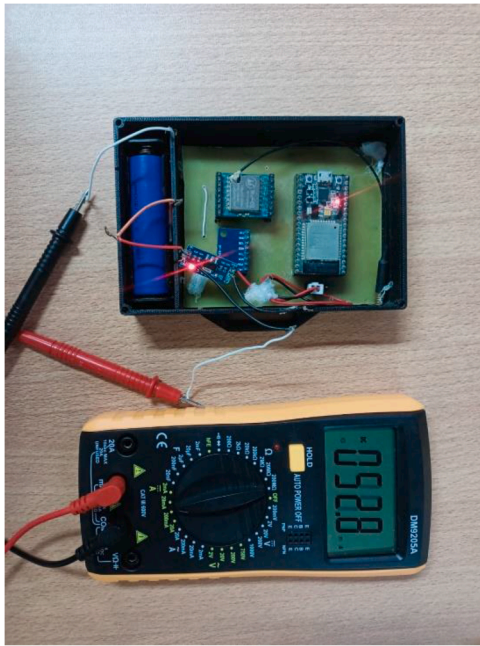


Fig. 8. Current measurement of the sensor node during normal operation (active mode: sensing and LoRa transmission).

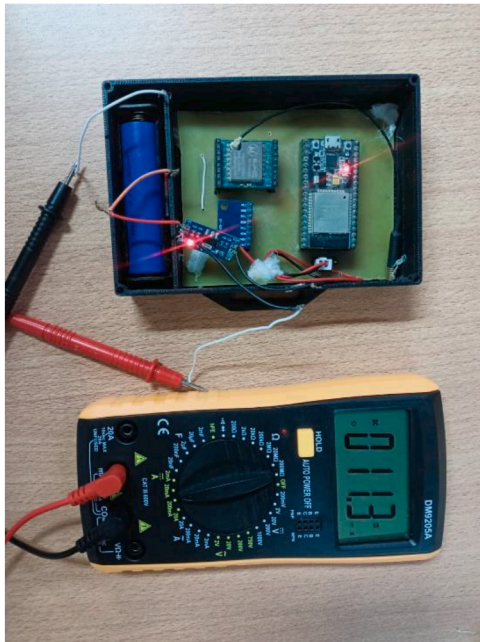


Fig. 9. Current measurement of the sensor node during deep sleep mode.

This result is consistent with the measured autonomy, confirming that the combination of deep sleep and the chosen duty-cycle ensures multi-day operation in field conditions.

3.6. LoRa communication setup

The performance of the LoRa wireless link is highly dependent on both the antenna design and the selection of communication parameters, particularly the Spreading Factor (SF), which directly affects transmission range, latency, and power consumption. To evaluate these aspects, two types of antennas were experimentally tested:

Table 1

System operating parameters and measured current values.

Parameter	Value	Description
Sensing interval	1 s	Accelerometer sampling period
Transmission interval (T _{cycle})	60 s	LoRa packet emission period
Battery capacity (C _{batt})	2200 mAh	Lithium-ion cell, nominal 3.7 V
Sleep current (I _{sleep})	11 mA	ESP32 light-sleep + AMS1117 quiescent
Sensing current (I _{sense})	2–5 mA	MPU-6050 acquisition
TX peak current (I _{TX})	52–53 mA	LoRa SX1278 transmission burst
Estimated average current (I _{avg})	21 mA	Duty-cycle weighted mean

- A standard helical spring antenna commonly supplied with SX1278 modules.
- A high-performance 433 MHz LoRa antenna (As shown in Fig. 10), used on both the transmitter and the receiver.

In addition, several SF values were assessed under both rural and urban open-field conditions to determine the most suitable configuration for reliable estrus detection in cattle. The LoRa transmission protocol was configured with the following baseline parameters to ensure robust and reproducible testing:

- Bandwidth (BW): 125 kHz.
- Coding Rate (CR): 4/5.
- Transmission power: 20 dBm.
- Frequency: 433 MHz.

Communication performance was experimentally evaluated in a rural environment at an approximate transmission distance of 200 m in order to assess the reliability of the proposed LoRa-based wearable system under real deployment conditions. During the tests, the received signal strength indicator (RSSI) values showed significant attenuation with increasing distance, with measured values ranging between approximately −151 dBm and −164 dBm. In addition, the packet reception ratio (PRR) was monitored to evaluate communication reliability, with an observed PRR of approximately 20% under the tested conditions.

The SNR is defined as the difference between the received signal strength indicator (RSSI) and the noise floor, as expressed in Eq. (4). Considering a typical LoRa receiver noise floor of approximately −120 dBm under the selected configuration

The signal-to-noise ratio (SNR) was estimated using the conventional relationship:

$$\text{SNR} = \text{RSSI} - \text{Noise Floor} \quad (4)$$

where the noise floor represents the background radio noise level of the communication channel. Considering a typical LoRa receiver noise floor of approximately −120 dBm under the selected configuration, the esti-



Fig. 10. High-performance LoRa 433 MHz antenna.

mated SNR values ranged between:

$$\text{SNR}_{\min} = -164 - (-120) = -44 \text{ dB} \quad (5)$$

and

$$\text{SNR}_{\max} = -151 - (-120) = -31 \text{ dB} \quad (6)$$

The estimated SNR values range between -44 dB and -31 dB , as shown in Eqs. (5) and (6). These results indicate challenging communication conditions, which contribute to variations in packet reception performance observed during the experiments. Nevertheless, the results confirm the feasibility of low-power long-range communication for livestock monitoring applications under real deployment conditions.

Despite these constraints, the experiments demonstrate the feasibility of long-range low-power communication for livestock monitoring applications.

3.7. System operation algorithms

A) Sensor Node Algorithm

The wearable sensor node operates using a low-power duty-cycling strategy designed to optimize battery lifetime while ensuring continuous activity monitoring. After system wake-up, the microcontroller initializes the LoRa communication module and the accelerometer sensor. The node then acquires acceleration data at a sampling frequency of 1 Hz (one sample per second). Between consecutive acquisitions, the microcontroller enters sleep mode to minimize energy consumption.

As shown in Fig. 11, after collecting 60 samples, the node locally processes the acquired data by calculating statistical features, including the mean, maximum, and minimum values of the acceleration axes. These aggregated parameters are subsequently transmitted through the LoRa communication interface to the gateway. Once the transmission process is completed, the system returns to sleep mode and repeats the acquisition cycle.

B) Gateway Algorithm

The gateway is responsible for continuously receiving, decoding, and

storing the data transmitted by the wearable sensor nodes. After system initialization and activation of the LoRa receiver, the gateway enters a permanent listening state awaiting incoming packets. When a valid packet is received, the transmitted data are decoded and processed before being automatically stored in an Excel-compatible logging file for monitoring and post-processing purposes. If no packet is detected, the gateway continues in reception mode until new data become available. As depicted in Fig. 12, This architecture ensures reliable real-time communication and centralized data collection while maintaining a lightweight implementation suitable for long-term deployment.

3.8. Web dashboard for monitoring and estrus detection

The developed web dashboard provides a unified interface for real-time monitoring, historical data exploration, and visualization of estrus detection results. Data are acquired at the gateway and transmitted to the dashboard via a UART-USB link, where they are served through a .JSON to the web interface. As illustrated in Fig. 13, The dashboard displays acceleration-versus-time plots for the three axes (X, Y, and Z), each shown in a dedicated graph. The graphical interface refreshes automatically every 1 min, synchronizing with the transmission interval of the sensor node. In addition to movement plots, the dashboard provides a real-time estrus detection status, automatically updating predictions while allowing users to filter data over a selected time range for historical analysis. Complementary information such as the last packet timestamp, node identifier, instantaneous acceleration magnitude, and link health status is also displayed, enabling a complete and interactive visualization of cattle activity.

The dashboard also includes a historical exploration mode. A date-range selector allows the user to specify either a predefined period (e.g., last hour, last day, last week) or custom start and end times. Upon selection, the backend queries the structured log file, stored in Excel (.xlsx) format and mirrored to an optimized cache, to return the corresponding time-bounded data. The historical chart retains the same interaction options as the live view, and users can export any selected period in CSV or Excel format for further analysis.

After data storage in the Excel file, the recorded information is automatically retrieved and displayed on the dashboard together with

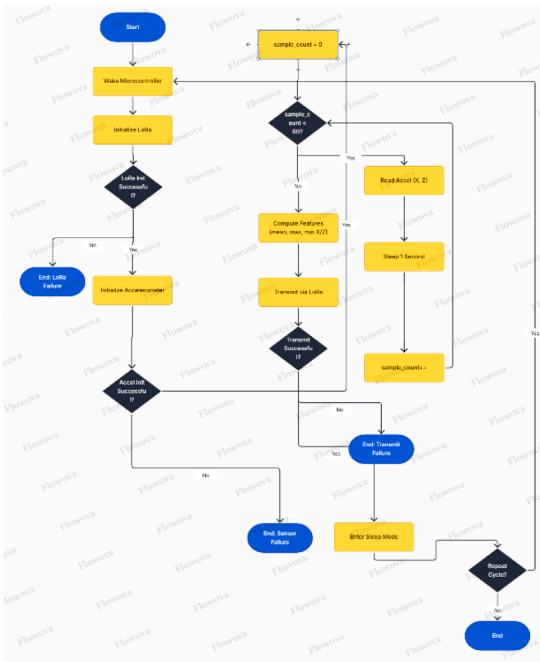


Fig. 11. Flowchart of the low-power sensor node algorithm for data acquisition, local processing, and LoRa transmission.

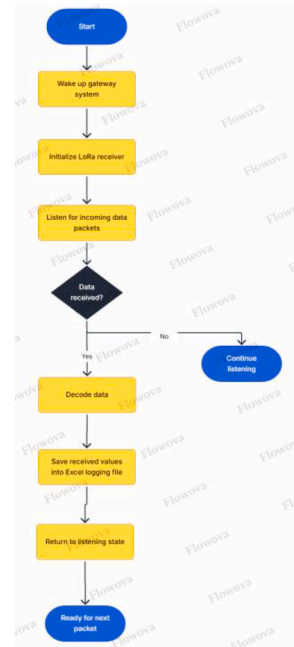


Fig. 12. Flowchart of the gateway algorithm for LoRa data reception and storage.

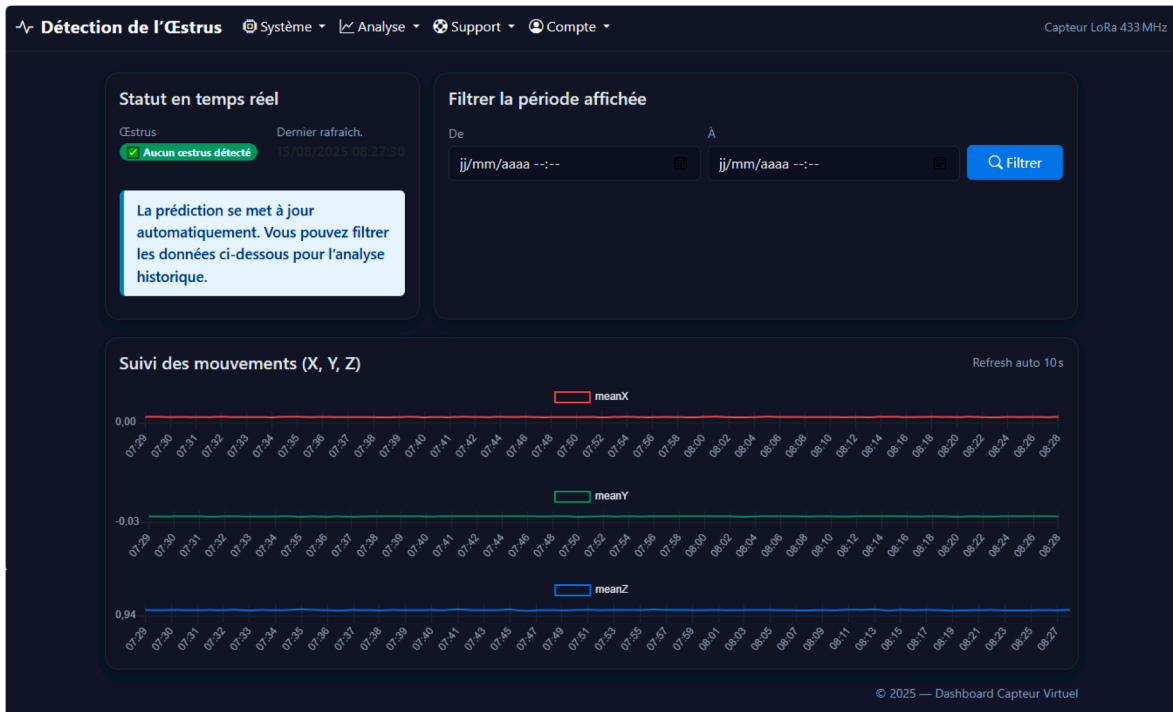


Fig. 13. Web dashboard for real-time and historical monitoring.

the processed results of the pre-trained model previously adopted from [46]. The model output, indicating the estimated estrus status with its associated confidence level, is shown alongside the raw acceleration data, allowing real-time visualization and comparison between sensor measurements and classification results.

3.9. Dataset structure and feature extraction

The collected sensor measurements were first recorded and stored in Excel (.xlsx) files for local backup and historical logging purposes. Subsequently, a dedicated Python-based data processing script was developed to automatically retrieve, preprocess, and transfer the recorded data into a structured MySQL database for long-term storage and analysis, as illustrated in Fig. 14. This automated pipeline enables efficient data organization, synchronization, and real-time accessibility through the monitoring dashboard.

For each 1-minute sliding window, several statistical features were extracted from the tri-axial accelerometer signals along the X, Y, and Z axes. The extracted features include maximum value (max), minimum value (min), mean value (mean), standard deviation (std), and differential variation (diff) calculated for each acceleration axis. The resulting dataset therefore contains the attributes max_x, min_x, mean_x, std_x, diff_x, max_y, min_y, mean_y, std_y, diff_y, max_z, min_z, mean_z, std_z, and diff_z, in addition to timestamp and sensor identification fields.

Estrus events were manually observed and annotated during the

experimental campaign in order to provide preliminary ground-truth labels associated with animal activity patterns and behavioral changes.

4. Results and discussion

The proposed system addresses key challenges in livestock monitoring by optimizing the electronic architecture, adopting efficient energy management strategies, and integrating long-range wireless communication. The results presented in this study focus on energy consumption and LoRa communication performance, including multiple experiments conducted with different spreading factors (SF), environments, and antenna configurations, as well as a comparative analysis with related works based on LoRa modules, sensors, and system implementation characteristics.

Different LoRa network modalities exist in the literature, including LoRaWAN, mesh networks, and point-to-point communication. LoRaWAN is based on a star-of-stars topology and provides scalability and centralized network management; however, it requires dedicated network infrastructure and introduces additional deployment complexity. Mesh LoRa networks can extend communication coverage but may increase energy consumption and communication overhead due to multi-hop transmissions. In contrast, the proposed system adopts a point-to-point LoRa architecture, which is more suitable for small-scale livestock monitoring applications, as it ensures low power consumption, simple deployment, and independence from external network

id	timestamp	max_x	min_x	mean_x	std_x	max_y	min_y	mean_y	std_y	max_z	min_z	mean_z	std_z	sensor_id
967	2025-04-03 07:21:57	3.32	7.94	6.4	3.03	0.27	2.88	9.59	2.06	0.31	5.28	9.7	1.89	0
966	2025-04-03 07:21:28	9.42	1.82	1.31	4.88	5.09	8.96	0.17	2.08	5.95	6.22	9.66	0.72	0
965	2025-04-03 07:20:27	1.76	0.68	4.33	2.3	5.83	2.54	5.64	0.02	6.51	9.94	6.39	3.44	0
964	2025-04-03 07:19:56	7.89	0.2	3.24	0.22	4.75	7.47	4.45	2.35	9.23	4.57	0.11	1.91	0
963	2025-04-03 07:19:27	9.09	0.34	5.11	2.04	5.29	7.86	6.76	3.95	0.56	9.82	7.8	4.42	0
962	2025-04-03 07:18:57	9.47	8.35	1.54	4.08	5.62	0.5	9.67	2.32	4.09	3.74	6.33	4.85	0
961	2025-04-03 07:18:26	6.74	1.97	6.75	3.16	0.35	1.54	5.13	2.22	5.28	6.51	3.94	3.04	0

Fig. 14. Structure of the collected accelerometer dataset stored in the MySQL database.

infrastructure.

The wearable sensor node was evaluated under real deployment conditions by mounting it on cows using a collar-based attachment system, as illustrated in Fig. 15. The device was designed to be lightweight and ergonomically suitable for long-term use, ensuring stable fixation without interfering with the animal's natural movement or behavior. Special attention was given to animal comfort during the design phase, including the reduction of sharp edges and the optimization of the enclosure shape. The system operates using low-power LoRa communication, which ensures minimal electromagnetic impact and is widely adopted in IoT applications. Overall, the deployment results indicate that the proposed system can operate effectively in real farm environments while maintaining animal welfare and providing reliable continuous monitoring.

In comparison with commercial livestock monitoring solutions such as Livecare Biocapsules, the proposed system offers several distinctive advantages. First, the device is designed as a detachable and reusable wearable collar, allowing easy transfer between animals, unlike implantable or proprietary fixed systems. Second, the proposed platform relies on an open hardware and firmware architecture, which facilitates customization, scalability, and reproducibility for research purposes as well as small-scale farming applications. Finally, the system is developed as a low-cost solution with a significantly reduced bill of materials compared to commercial alternatives, making it more accessible and suitable for deployment in resource-constrained environments.

In order to assess the practicality of the system, particular attention was given to the power supply design and its influence on energy consumption and operating autonomy. Several configurations were tested to quantify their effect on system performance. The use of a 7805 linear regulator significantly reduced operating time to less than one hour due to power losses. In contrast, a single 3.7 V Li-ion battery combined with a low-dropout regulator (AMS1117) extended the autonomy to approximately 104 h (four full days of continuous operation), with LoRa transmissions every 60 s and the ESP32 operating in deep sleep between events. These results highlight the critical role of power circuitry design in determining autonomy. The final choice of a 3.3 V regulated supply ensured a practical balance between simplicity and efficiency, enabling

multi-day operation under real farm conditions. Beyond estrus detection, these findings demonstrate principles applicable to a wide range of IoT agricultural devices where energy efficiency is essential.

The evaluation of LoRa communication confirmed that antenna selection and spreading factor configuration are decisive for ensuring reliable transmission. Table 2 summarizes the LoRa communication ranges under different conditions. The high-performance 433 MHz antenna substantially improved the range compared to the standard spring antenna, reaching up to 200 m in open rural environments at SF12. However, in urban areas with obstacles, the range dropped below 100 m. Increasing the spreading factor allowed longer distances but introduced latency and higher energy consumption, illustrating the trade-off between coverage and efficiency. These results emphasize that hardware selection and parameter tuning are equally important to system reliability and are representative of the challenges faced by IoT networks deployed in heterogeneous farm environments.

Table 3 provides a comparative overview of the proposed system and other recent LoRa-based implementations reported in the literature, covering livestock monitoring as well as additional application areas.

Table 2

Experimental LoRa communication range under different spreading factors and antenna configurations.

Spreading Factor (SF)	Environment	Antenna Type	Communication Range
SF7	Rural	Standard spring antenna (433 MHz)	Approx. 10 m
SF9–SF12	Rural	Standard spring antenna (433 MHz)	Between 30 and 50 m
SF7	Rural	High-performance antenna (433 MHz)	Approx. 50 m
SF10	Rural	High-performance antenna (433 MHz)	Between 100 m and < 200 m
SF12	Rural	High-performance antenna (433 MHz)	Approx. 200 m (with higher delay and power consumption)
SF7–SF12	Urban	High-performance antenna (433 MHz)	< 100 m (maximum)



Fig. 15. Deployment of the wearable sensor node on a cow using a collar-based attachment system.

Table 3

Comparative analysis of LoRa-based systems reported in the literature and the proposed work.

Reference	Sensors	LoRa Module	Cost	Battery Life	Range	energy per transmission	gateway type
Salah et al. (2022) [47]	Accelerometer	RFM95W	-	53 h	180 m	37.72 mA	Commercial
Rajashekhar et al. (2024) [48]	PIR sensors, GPS technology	Rylr998	\$131	-	120 m	-	in-house
Xie et al. (2022) [49]	Laser Distance Meter (Disto E7300)	SX1276 & USRP X310	-	-	20 m	-	Commercial
Marco et al. (2017) [51]	temperature and humidity sensor	RFM95	-	75 min	115 m	-	-
B Setiawan et al. (2019) [52]	-	SX1278	-	-	200 m	-	-
Hu et al. [8].	Commercial robot	SX1280	-	-	200 m	-	in-house
Jiang et al. (2021) [50]	Backscatter	SX1276	-	-	100 m	-	in-house
This work	Accelerometer	SX1278	\$120	4 days	200 m	11 mA, 53mA	in-house

The analysis shows that many existing works do not explicitly report battery life, making it difficult to assess their applicability in long-term deployments. Among those that do, autonomy rarely exceeds two days, whereas the proposed system sustains operation for four consecutive days, representing a significant improvement in energy efficiency. Regarding communication range, previous studies report values between 20 m and 200 m depending on the environment and hardware used. Our system consistently achieved 200 m in rural conditions, which is competitive given the constraints of the SX1278 module and confirms the robustness of the design. Beyond estrus detection, these results illustrate general principles in balancing autonomy and connectivity for IoT systems in agriculture.

The average power consumption of the proposed wearable node was estimated based on the measured current in both sleep and transmission modes, operating at a supply voltage of 3.3 V. Using the duty-cycling configuration of the system, the transmission phase (53 mA) occurs briefly every cycle, while the node remains in sleep mode (11 mA) for the majority of the time. Based on a 60-second operating cycle with a short transmission duration of approximately 0.2 s, the resulting average power consumption is approximately 36.8 mW. This value reflects the effectiveness of the proposed energy management strategy, which significantly reduces power usage by limiting high-current transmission periods and maximizing low-power sleep operation, thereby improving the autonomy of the wearable device.

Taken together, the results demonstrate that the proposed system achieves a robust balance between energy autonomy and communication reliability, two key requirements for continuous livestock monitoring. While the autonomy is already superior to most comparable systems, future improvements could target further range extension in urban environments. More broadly, the findings are not limited to estrus detection but can be generalized to IoT-based agricultural devices that face similar constraints in energy management and long-range communication.

5. Limitations and future work

The limitations of the proposed system are clearly acknowledged in this study. The current work mainly focuses on energy optimization and communication performance evaluation; therefore, large-scale real animal testing has not yet been conducted. In addition, the measured LoRa communication range using SX1278 (RA-02) modules was approximately 200 m under the tested conditions. Future work will focus on enhancing the communication range through optimized gateway deployment or advanced LoRa modules, as well as further improving device robustness for harsh agricultural conditions.

6. Conclusion

This work presented a comprehensive approach to estrus detection in dairy cows by addressing not only classification algorithms but also the underlying electronic and communication architecture. The developed system successfully combines low-power hardware design, efficient energy management, and LoRa-based communication to achieve autonomous and reliable operation in realistic farming conditions.

Experimental results confirmed a battery autonomy of up to four days of continuous operation and a stable communication range of approximately 200 m. Compared to existing solutions, the proposed design demonstrates improved energy efficiency, extended battery discharge duration, and reliable data transmission with minimal infrastructure requirements. These findings underline the importance of integrates optimized hardware design with intelligent energy management strategies in the context of precision livestock farming.

Ethical statement

This research did not involve human participants or animals. Therefore, ethical approval and informed consent were not required.

CRedit authorship contribution statement

Hechmi Salhi: Writing – original draft. **Houda Jouini:** Writing – review & editing. **Hatem Allagui:** Writing – review & editing. **Abdelkader Mami:** Validation.

Declaration of competing interest

The authors declare that there is no conflict of interest.

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Data availability

No data was used for the research described in the article.

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